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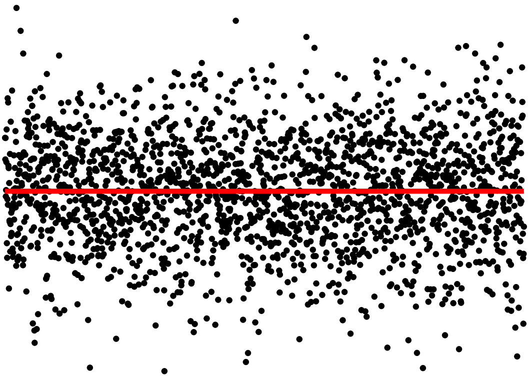
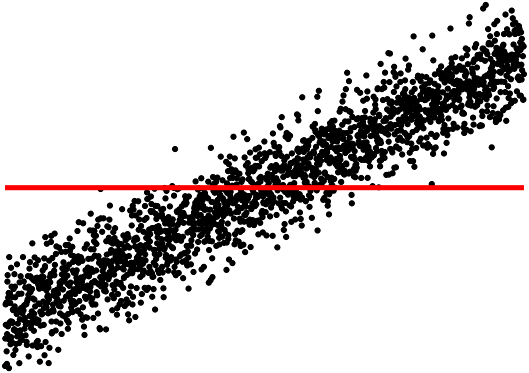
Tackling Global Issues: Change

Rebecca Killick(r.killick@lancs.ac.uk)
CASI 2024

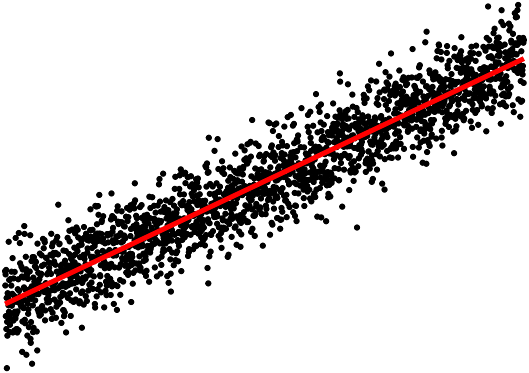
Model development



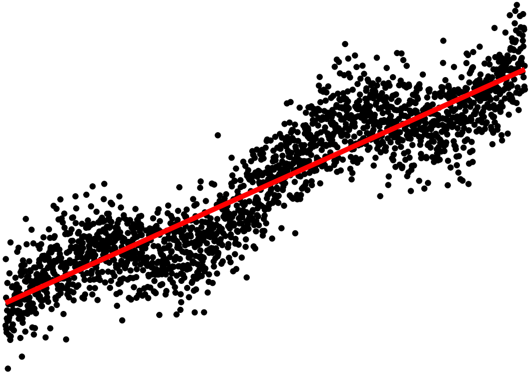
Model development



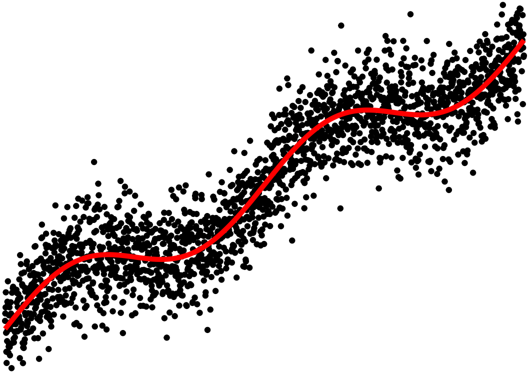
Model development



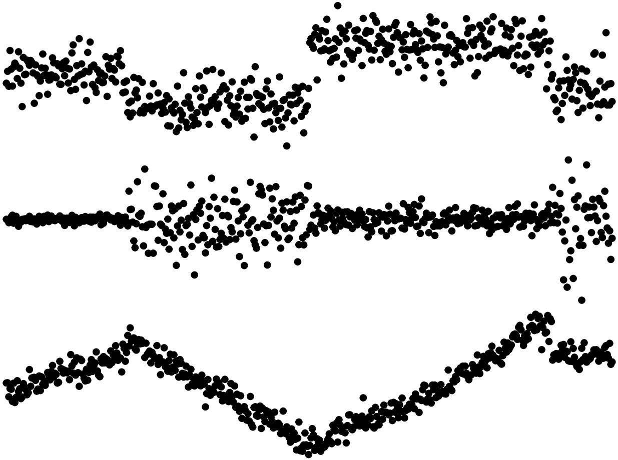
Model development



Model development



Model development



AIM

To convince you that there are many interesting problems and challenges and to invite you to the join the cause!

Outline

- What are changepoints?
- Tour of changepoint problems and applications
    - Speed & accuracy
    - Autocorrelation
    - Model choice
    - Multivariate
- Open problems
    - Autocorrelation
    - Multivariate
    - Sensitivity to data
    - Online inference

What are changepoints?

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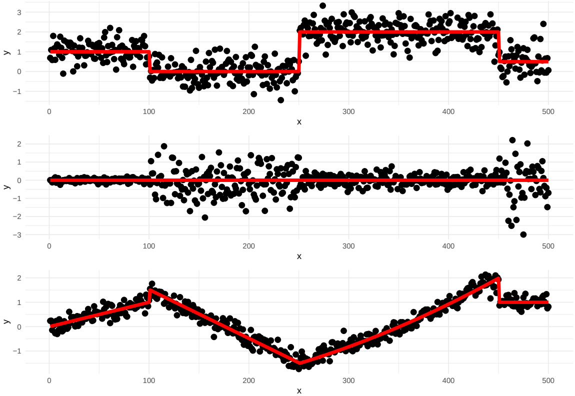
For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

There are many different types of change.



Problem

- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

Interesting Challenges

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- Optimization
- Model development
- Theoretical properties
- Limits of detection
- Small sample and large sample in the same data

Speed / Accuracy

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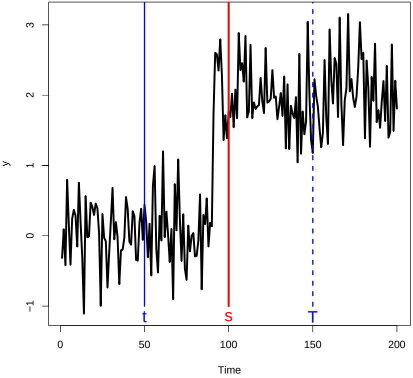
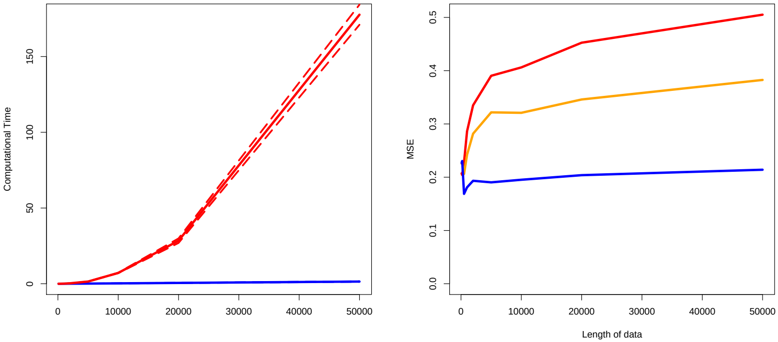
Computational complexity in 2010: $\mathcal{O}(n^2)$ exact, $\mathcal{O}(nlogn)$ approx.

$\mathcal{O}(n^2)$ solution book keeps the last changepoint location. BUT . . .



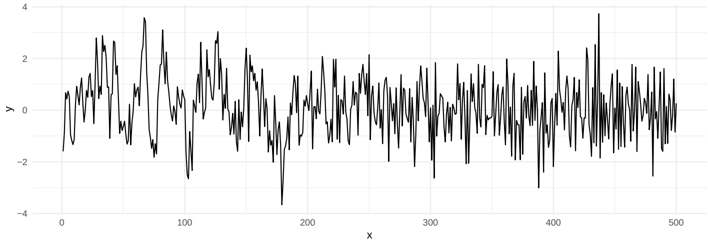
Speed / Accuracy

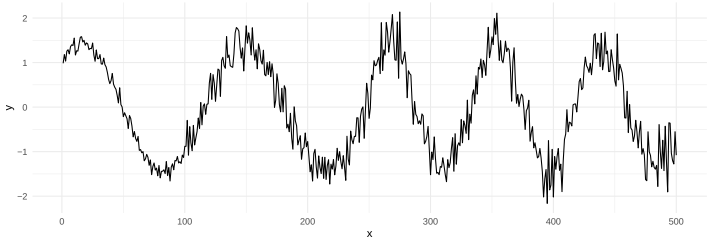
. . . resulting in an $\mathcal{O}(n)$ algorithm if changepoints occur regularly.



SPEED + ACCURACY = PELT!

More complicated changes





Is climate change speeding up? Here's what the science says.

This year's record temperatures have some scientists concerned that the pace of warming may be accelerating. But not everyone agrees.

By Chris Mooney and Shannon Osaka

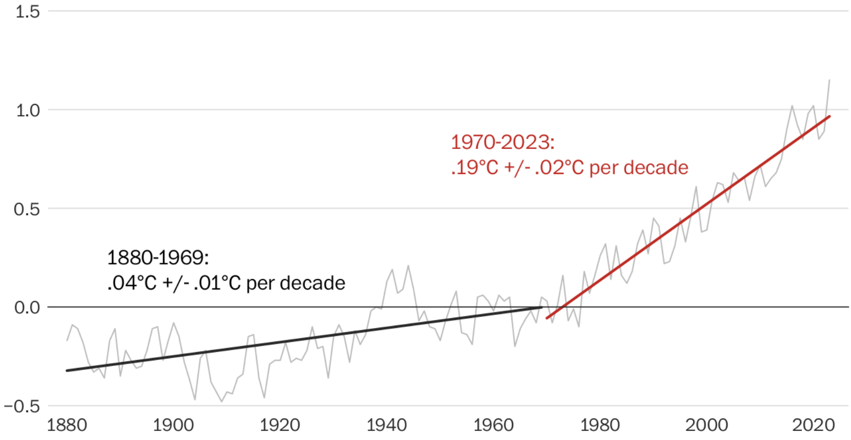
December 26, 2023 at 6:30 a.m. EST

For the past several years, a small group of scientists has warned that sometime early this century, the rate of global warming — which has remained largely steady for decades — might accelerate. Temperatures could rise higher, faster. The drumbeat of weather disasters may become more insistent.

Where is Statistics?

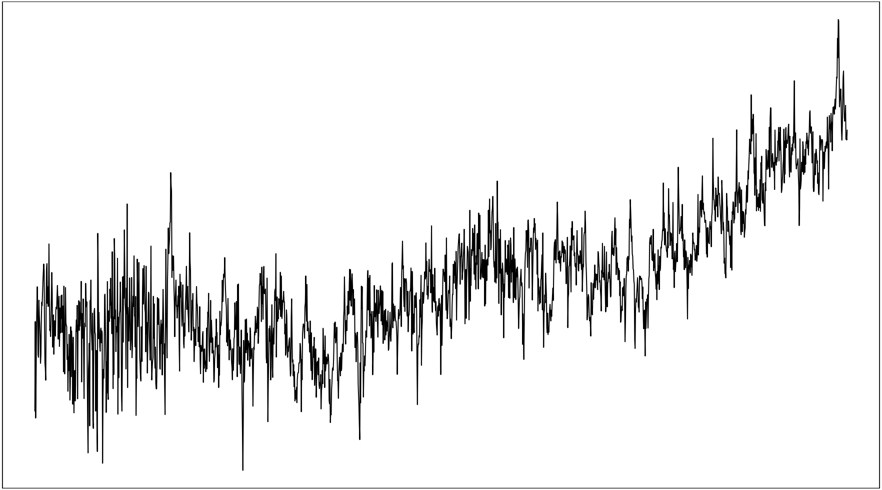
The increased rate of global warming

Values are relative to the 1951-1980 global mean temperature, in degrees Celsius



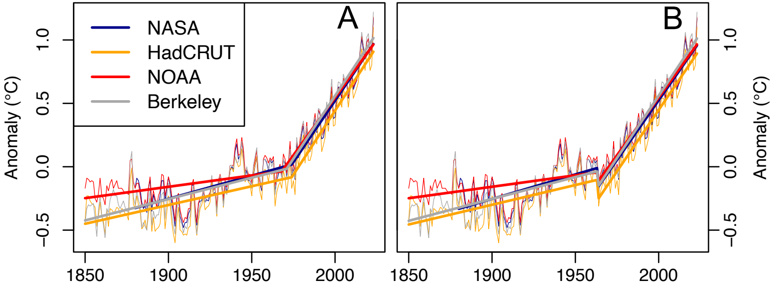
Note: 2023 is an estimate based on values from January through November.

Natural correlation



Climate Change

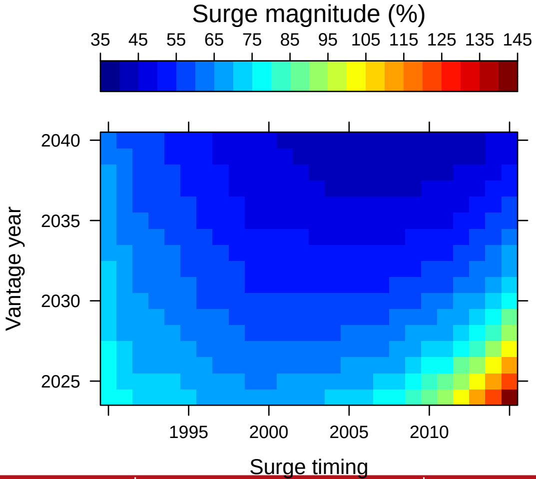
- Has there been a recent surge in global warming?
- Analysed several surface temperature time series estimates.



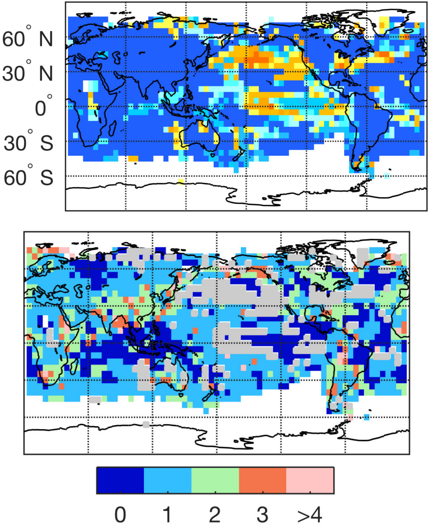
- Considered all possible models prevalent in the literature
- None show changes in the recent data

So what?

What would we need to see (in the statistically preferred model).

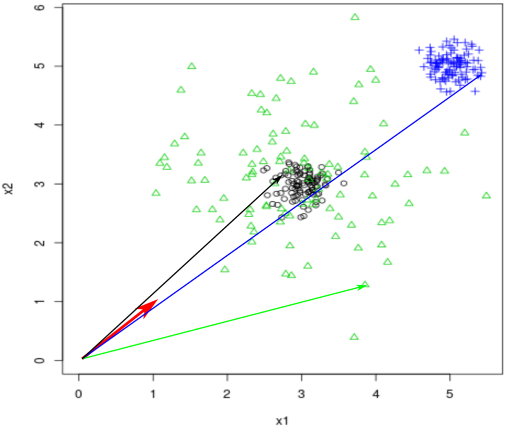


Climate Dynamics



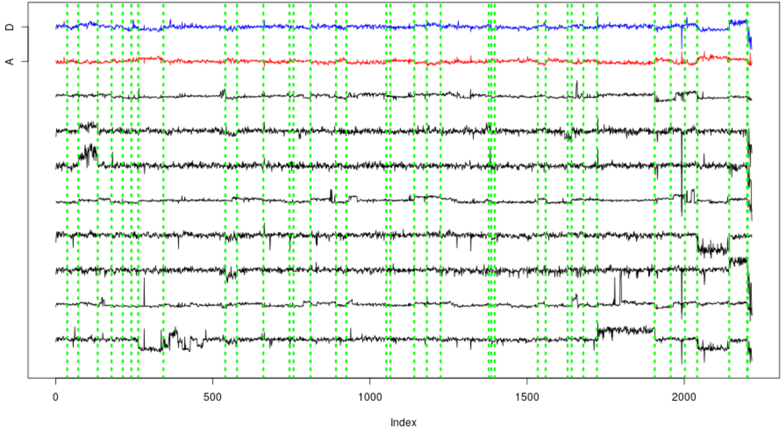
Health

- Which genomic markers could be indications for disease?
- Mean and variance changes



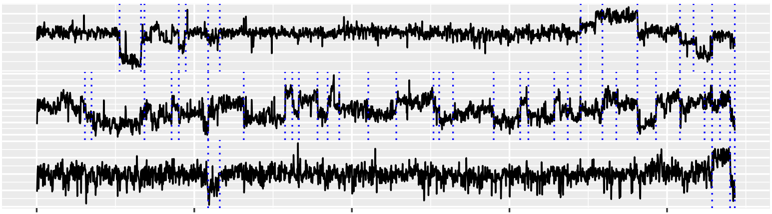
Health

- Which genomic markers could be indications for disease?
- Use two transformations instead of one



Health

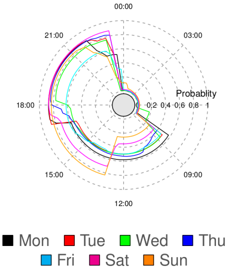
- Which genomic markers could be indications for disease?
- Only markers present in lots of patients are worth investigating.

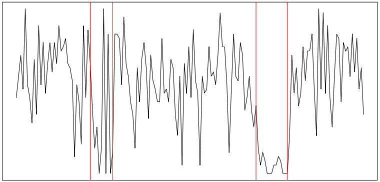


- Developed computationally efficient subset multivariate changepoint methodology.
- Only 10 changes are present in more than 60% of the patients.
- Over 50 detected by competitors.

OP: Aging

- Passively monitor patients at home
- Desire to have early intervention before crisis point is reached





- View data as circular, e.g., each day.
- Developed circular changepoint detection to detect changes in activity levels within a day
- Changes in the detected routine signal early intervention

OP: Sensitivity

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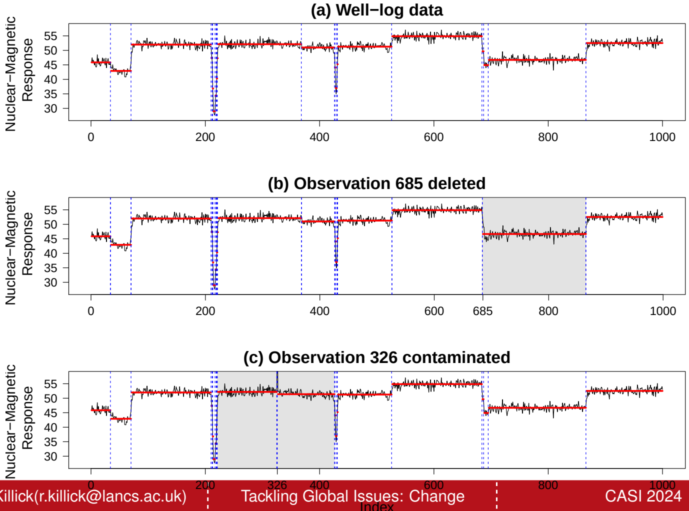
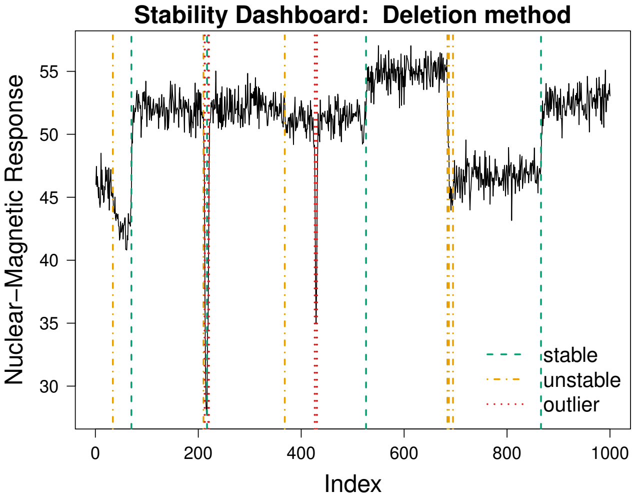
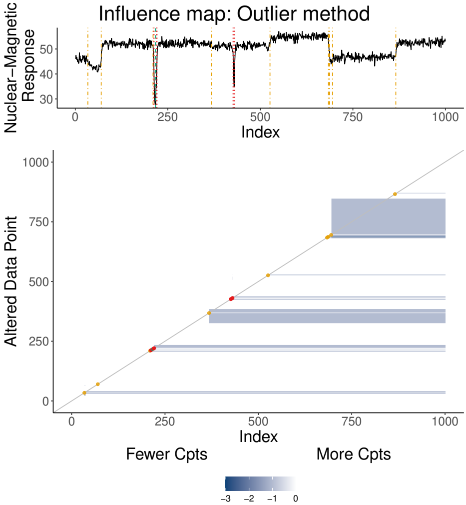
- Changepoint locations are sensitive to the observed data
- Some changepoint algorithms are more sensitive than others.



OP:Sensitivity



OP:Sensitivity



Open Problems

- Optimization
- Autocorrelation
- Multivariate data
- Model complexity
- Online changes

Summary

- Changepoints occur in most data
- Detecting and documenting them improves analyses
- Multivariate, online, big data mostly isn't a problem
- But there are still plenty of open problems in the field
- I enjoy working with messy real life data . . .
- . . . as often it sparks my next research challenge.

Papers discussed

Preprints of all available at: www.lancs.ac.uk/~killick/pub.html

PELT: https://doi.org/10.1080/01621459.2012.737745

Model Choice: https://doi.org/10.1175/JCLI-D-17-0863.1 & https://doi.org/10.1002/qre.2712

LMvsCpts: https://doi.org/10.1007/s11222-017-9731-0 & https://doi.org/10.1002/env.2568

Multivariate: https://doi.org/10.1038/s41598-019-46836-y & https://doi.org/10.1007/s11222-020-09940-y

Autocorrelation: https://doi.org/10.1002/sta4.351

Recent papers

Circular time: http://doi.org/10.1111/rssc.12472

Sensitivity: https://doi.org/10.1080/10618600.2021.2000873

Multivariate: https://arxiv.org/abs/2108.07340

Autocorrelation: https://arxiv.org/abs/2102.10669

Warming Surge: https://arxiv.org/abs/2403.03388

Online: Email for preprint, code:
https://github.com/grundy95/changepoint.forecast

Autocorrelation Change

Change in variance with nonstationary errors:
https://doi.org/10.1002/env.2576

Changes in second order structure within nonstationary series:
http://dx.doi.org/10.1214/13-EJS799

Uncertainty of autocovariance changes:
https://doi.org/10.1080/00401706.2014.902776